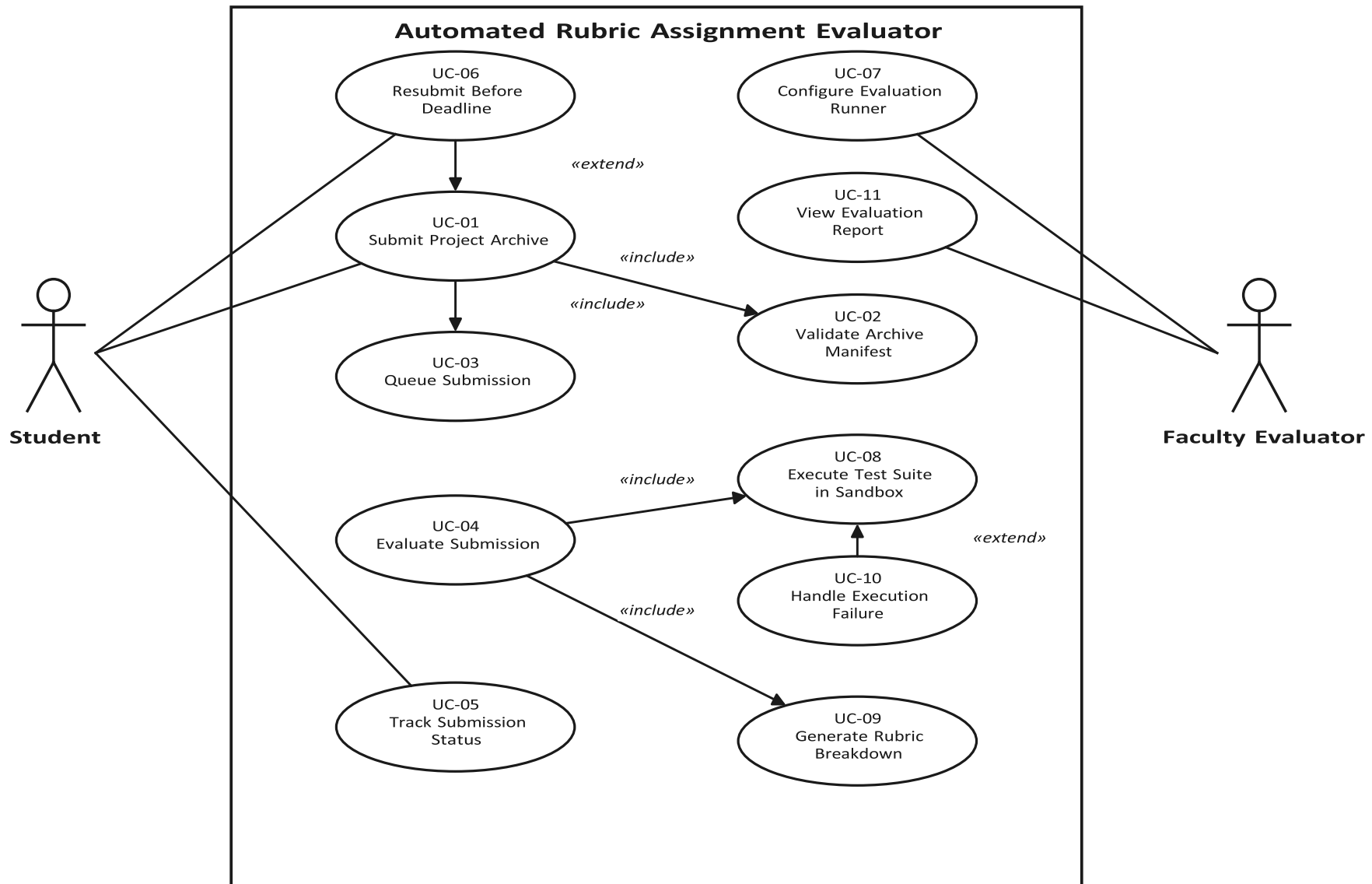


## Requirements Table — Test Case 1: Submission Intake & Automated Evaluation Pipeline

| ID      | Type                                        | Description                                                                                                                                                                                                                                                          | Priority | Acceptance Criteria                                                                                                                                                                                                                                                                        | Rationale                                                                                                  |
|---------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| FR-001  | Functional — Submission Intake              | The system shall allow a Student to upload a project archive (.zip, ≤ 50 MB) against an open assignment, validate the archive structure against the assignment manifest, and register the submission with a unique submission ID, SHA-256 hash and server timestamp. | High     | <b>Pass:</b> a manifest-conformant archive is accepted and a submission ID with hash and timestamp is returned; a missing required file is rejected with the exact path named.<br><b>Fail:</b> a malformed archive is accepted, or an accepted archive has no hash recorded.               | The manifest check catches the most common student error early and prevents wasted evaluation runs.        |
| FR-002  | Functional — Evaluation Queue               | The system shall place every accepted submission on a FIFO evaluation queue, dispatch it to an isolated runner as capacity frees, and retry a run at most twice on infrastructure failure before marking it failed.                                                  | High     | <b>Pass:</b> submissions are dispatched in arrival order, queue position is visible to the Student, and an infrastructure-failed run is retried and then marked failed on the third attempt.<br><b>Fail:</b> a queue item is silently dropped or starved indefinitely.                     | Batch evaluation only works if load is smoothed; unbounded retries would hide genuine defects.             |
| FR-003  | Functional — Automated Test Execution       | The system shall compile each submission and execute the assignment's pre-configured unit test suite inside a sandbox, recording per-test outcome, execution time and captured stdout/stderr.                                                                        | High     | <b>Pass:</b> for a submission with 40 tests, all 40 outcomes plus logs are persisted and linked to the submission.<br><b>Fail:</b> partial test results are stored, or logs are discarded on failure.                                                                                      | The test outcome record is the evidence base for every score and every later appeal.                       |
| FR-004  | Functional — Rubric Score Breakdown         | The system shall map each test outcome to its bound rubric criterion, compute marks per criterion using the published rubric weights, and render an itemized breakdown whose total equals the sum of criterion marks.                                                | High     | <b>Pass:</b> the rendered breakdown lists every criterion with marks awarded, tests passed/total, and a total matching the sum, within 10 s of submission.<br><b>Fail:</b> a criterion is unscored, or the total disagrees with the itemized marks.                                        | An itemized breakdown is what makes an automated score defensible to the student.                          |
| FR-005  | Functional — Status Tracking & Resubmission | The system shall expose submission status (QUEUED, RUNNING, EVALUATED, FAILED) with live queue position, notify the Student on completion, and accept resubmissions until the deadline while retaining every earlier version and treating the latest as current.     | Medium   | <b>Pass:</b> status transitions are visible without a page reload; a resubmission supersedes the previous score while all earlier versions stay retrievable; a submission after the deadline is refused.<br><b>Fail:</b> version history is overwritten, or a late submission is accepted. | Students iterate against the autograder, and the version trail is needed for plagiarism and appeal review. |
| NFR-001 | Performance & Security                      | The system shall handle up to 100 concurrent project submissions without dropping queue items or exceeding a 1.5 GB memory footprint, holding 95th-percentile evaluation latency at or below 10 s per submission.                                                    | High     | <b>Pass:</b> a load benchmark at 100 concurrent submissions records zero dropped items, peak RSS ≤ 1.5 GB and p95 latency ≤ 10 s.<br><b>Fail:</b> any dropped item, memory breach or latency above target.                                                                                 | An entire cohort submits in the final hour before a deadline, so peak load is the design point.            |
| NFR-002 | Security & Reliability (Sandbox Isolation)  | Student code shall run in a network-isolated sandbox with a read-only test harness mount, a 2-core CPU limit and a 60 s wall-clock timeout; a runner crash shall not affect other queued submissions.                                                                | High     | <b>Pass:</b> a submission attempting network egress, writes outside its workspace, or an infinite loop is terminated and logged while other jobs complete normally.<br><b>Fail:</b> sandbox escape, or one submission stalling the queue.                                                  | The system executes untrusted code by design, making the runner the primary attack and failure surface.    |



— association

→ «include» / «extend» dependency

Test Case 1 \_ Submission Intake & Automated Evaluation Pipeline

## Use-Case Flow Specification — Test Case 1

|                      |                                                                       |
|----------------------|-----------------------------------------------------------------------|
| Use Case ID          | UC-04                                                                 |
| Use Case Name        | Evaluate Submission (automated rubric evaluation)                     |
| Primary Actor        | Student                                                               |
| Secondary Actors     | Faculty Evaluator (owns rubric and test suite); sandboxed test runner |
| Trigger              | A submission reaches the head of the evaluation queue                 |
| Related Requirements | FR-002, FR-003, FR-004, NFR-001, NFR-002                              |
| Included Use Cases   | UC-08 Execute Test Suite in Sandbox, UC-09 Generate Rubric Breakdown  |

### Preconditions

- The Student is authenticated and enrolled in the course offering.
- The submission has been accepted and manifest-validated (UC-01, UC-02).
- A rubric version and a bound unit test suite are published for the assignment.
- At least one evaluation runner has free capacity.

### Postconditions — success guarantee

- An execution report with per-test outcomes, timings and logs is persisted.
- An itemized rubric breakdown is stored against the submission and is visible to the Student and the Faculty Evaluator.
- Submission status is set to EVALUATED and the Student is notified.
- An audit entry records the rubric version and test-suite version used.

### Postconditions — minimal guarantee

- No criterion marks are persisted; status is set to FAILED and the previous evaluated version, if any, remains the current score.

### Main Success Scenario

1. The system dequeues the submission and provisions a network-isolated sandbox.

2. The system mounts the test harness read-only and applies the CPU and wall-clock limits.
3. The system unpacks the archive into the sandbox workspace and compiles the project.
4. The system executes the published unit test suite and captures per-test outcome, duration and output — «*include*» UC-08.
5. The system maps each test outcome to its bound rubric criterion.
6. The system computes marks per criterion from the published weights and totals them — «*include*» UC-09.
7. The system destroys the sandbox and persists the execution log and the itemized breakdown.
8. The system sets submission status to EVALUATED and notifies the Student.
9. The Student opens the submission page and views the breakdown with per-test evidence.

**Alternate Flow A1 — compilation failure, timeout or sandbox violation**

1. At step 3 or 4 compilation fails, a test exceeds the 60 s wall-clock limit, or the sandbox terminates the process for attempted network access.
2. The system halts execution, destroys the sandbox and captures the compiler output or termination reason.
3. The system sets status to FAILED, writes the reason to the audit log and persists no criterion marks.
4. The system notifies the Student with the captured diagnostics and flags the submission in the Faculty Evaluator's manual-review queue.
5. If the deadline has not passed the Student resubmits and the use case restarts at step 1; otherwise the Faculty Evaluator resolves the submission manually.